

Lemma 1.5 $\Leftrightarrow$ ab then

$P \xrightarrow{\varphi} L$ L simple $\varphi \neq 0$
 is a proj cover $\Leftrightarrow$ P has a
 max proper subobj $\text{Rad } P$ $\Leftrightarrow \text{End}_A(P)$
 local

Proof: "1 $\Rightarrow$ 2" $\ker \varphi$ is max (is this claim or fact?)

Assume $U \subset P$ subobj $U \not\subset \ker \varphi$

Then $P \xrightarrow{\varphi} L$ simple $\varphi \text{ ess.}$ $\Rightarrow U \xrightarrow{\sim} P$
 $\uparrow \nearrow$
 $U' \neq 0$

"2 $\Rightarrow$ 3" Note: P indecomposable (why?)

Claim: $f \in \text{End}_A(P)$ then $f \text{ epi} \Leftrightarrow f \text{ iso}$

(Indeed $\ker f \hookrightarrow P \xrightarrow{f} P$ splits P proj but
 P is indecomp $\Rightarrow \ker f = 0$)

Let $\alpha, \beta \in \text{End}_A(P)$ not iso

Claim: α, β not epi $\Rightarrow \text{im}(\alpha + \beta) \subset \text{im}(\alpha) + \text{im}(\beta) \subset \text{Rad } P$

$\Rightarrow \alpha + \beta$ not epi $\Rightarrow$ not iso $\Rightarrow$ max ideal.

"3 $\Rightarrow$ 1" Assume

$P \xrightarrow{\varphi} L$ If $\alpha \in \text{Rad}(\text{End}_A(P))$
 $\alpha \uparrow \nearrow$
 $P \xrightarrow{\varphi}$ Nakayama $\Rightarrow$ Contradiction

$\Rightarrow \alpha$ is invertible $\Rightarrow P \xrightarrow{\varphi} L$ proj cover $\blacksquare$

Thm 1.3 $\mathcal{A}$ ab fl & has enough projectives

$$\{ \text{indec proj obj} \} / \cong \xleftrightarrow{1-1} \{ \text{simple obj} \} / \cong$$

$$P \mapsto \text{unique indec quot}$$

$$\text{proj cover } P_L \xleftarrow{1-1} L$$

Proof: Assume that proj covers exist (a bit later)

$$L \mapsto P_L \xrightarrow{\varphi} L \text{ unique up to iso}$$

P_L is indecomposable by proof of Lemma 1.5

$\Rightarrow$ well-defined ($\leftarrow$)

If P indecomp. proj then $\exists L$ simple $P \twoheadrightarrow L$

since $\mathcal{A}$ fl

$$\begin{array}{ccc} P & \twoheadrightarrow & L \\ \downarrow \varphi & \nearrow & \\ P_L & & \end{array} \xRightarrow{\text{by ess}} P \twoheadrightarrow P_L$$

what is this argument?

Since P_L proj $\Rightarrow$ surj splits $\xRightarrow{P \text{ indec}} P \cong P_L$

$\Rightarrow (\mapsto)$ is well defined and furthermore

$\mapsto$ and $\leftarrow$ are inverses by construction

Now, we prove the existence of proj covers

[Prop 1.6] R ring TFAE

- I) Every fg R -module has a proj cover
 - II) Every simple R -module has a proj cover
 - III) $R = P_1 \oplus \dots \oplus P_r$ for some indec(proj) R -modules
 - IV) fg proj objs form Krull-Schmidt category
(then R is called semiperfect)
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[Def] An additive category is Krull-Schmidt

$\Leftrightarrow$ every $X \in \mathcal{C}$ can be written as $X \cong X_1 \oplus \dots \oplus X_r$
with $\text{End}_{\mathcal{C}}(X_i)$ local

Proof: left as an exercise

Idea: Every obj in an ab cat can be
"realised" as fg proj object in a module category

[Lemma 1.7] $\mathcal{C}$ additive cat $X \in \mathcal{C}$ $R := \text{End}_{\mathcal{C}}(X)$

then $\text{Hom}_{\mathcal{C}}(X, -) : \mathcal{C} \rightarrow R^{\text{op}}\text{-mod}$ gives

fully faithful functor

$\text{Add}(X) := \langle X \rangle_{\oplus}$ Cat generated by X wrt $\oplus, \otimes$
 $\downarrow$
 $\text{Proj}(R) := \text{fg proj } R^{\text{op}} \text{ modules}$

It is an equivalence if $\text{End}_C(X)$ has split idempotents

(i.e. if $\varphi \in \text{End}_C(X)$ $X \xrightarrow{\pi} Y \xrightarrow{i} X$ then $\pi \circ i = \text{id}_Y$)

Proof: $\text{Hom}_C(X, X) \xrightarrow{\sim} \text{Hom}_{-R}(FX, FX)$ is iso
since $FX = R$. Since the functor is additive
it's compatible w/ $\oplus$, also w/ summands $\oplus$ by the 5-lemma

If split idempotents then functor dense.

Note $\text{Proj}(R) = \langle R \rangle_{\oplus} = \text{Add}(R)$

Consequence: Cor 1.8 An additive category is

Krull-Schmidt $\Leftrightarrow C$ has split idempotents and

$\text{End}_R(X)$ is semiperfect $\forall X \in C$ □

Want: Tools to construct Krull-Schmidt categories

$\Rightarrow$ bichain by Atiyah

Def at ab, a bichain is a collection of morphisms

$$X_n \xrightarrow{\alpha_n} X_{n+1} \xrightarrow{\beta_n} X_n \quad n \in \mathbb{N}_0$$

$X \in \mathcal{A}$ satisfies bichain condition if

$\forall$ bichain satisfies $X_0 = X \leadsto \alpha_n \& \beta_n$ invertible $n \gg 0$

Lemma 1.9 Any obj X in a fl ab cat satisfies bichain condition

Proof: Consider $\ker(\alpha_r - \alpha_0)$ and $\operatorname{im}(\beta_1 - \beta_r)$
 $\uparrow$ $\uparrow$
asc chain desc chain

Since X has fl it must stabilise $\Rightarrow \alpha_r$ and β_r iso for sufficiently large r $\square$

Lemma 1.10 Any obj in a k -linear Hom-finite cat satisfies the bichain condition $\hookrightarrow \dim \operatorname{Hom}(-, -) < \infty$

Proof: $X_n \twoheadrightarrow X_{n+1} \hookrightarrow X_n$ w/ $X_0 = X$

$$\operatorname{Hom}(X_{n+1}, X_{n+1}) \xrightarrow{\beta_0} \operatorname{Hom}(X_{n+1}, X_n) \xrightarrow{\alpha} \operatorname{Hom}(X_n, X_n)$$

Since Hom is left exact β_0 is mono

(I don't exactly understand from here)

if β_0 bijective then β is epi

if α bijective then α is mono

But by assumption β_0, α must be bijective

(at some point $(n \gg 0)$)

$\rightarrow$ which assumption?

Lemma 1.11 (Fitting lemma)

$X \in \mathcal{A}$ satisfying bichain condition, $\varphi \in \text{End}_{\mathcal{A}}(X)$ then

$$X = \text{im } \varphi^r \oplus \ker \varphi^r \text{ for some } r \quad \left(\begin{array}{l} \text{isn't it } r^* \text{ from the?} \\ \text{is bichain needed for} \\ \text{the homomorphism then?} \end{array} \right)$$

In particular, $X \text{ indec} \Rightarrow \varphi \text{ nilp/iso}$.

Proof Let $X_n \xrightarrow{\alpha_n} X_{n+1} \xrightarrow{\beta_n} X_n$ with $X_0 = X$, $X_n = \text{im } \varphi^n$

then $\alpha_n = \varphi$, $\beta_n = \text{id}$. This is a bichain and by

assumption $\text{im } \varphi^r \xrightarrow{\varphi} \text{im } \varphi^{r+1}$ for some $r \gg 0$

then $\varphi^r: \text{im } \varphi^r \rightarrow \text{im } \varphi^{2r}$ $r \gg 0$ with inverse φ^r

$$\text{im } \varphi^r \oplus \ker \varphi^r \xrightarrow{(\alpha_1, \alpha_2)} X$$

Let $\pi_1 = \varphi^{2r}: X \rightarrow \text{im } \varphi^r$; $\pi_2 = \text{id} - \varphi^{2r}: X \rightarrow \ker \varphi^r$

but $\alpha_1 \circ \pi_1 + \alpha_2 \circ \pi_2 = \text{id} \Rightarrow X = \text{im } \varphi^r \oplus \ker \varphi^r \quad \square$

Lemma 1.12 $X \in \mathcal{A}$ satisfying bichain condition

then $X \text{ indec} \Leftrightarrow \text{endo ring is local}$

Proof: Assume $X = X_1 \oplus X_2$, $\text{id} = \varepsilon_1 + \varepsilon_2$, ε_i - idempotent

but the sum is invertible $\Rightarrow$ contradiction

to max ideal should consist of non-inv morphisms

the remaining direction is an exercise

$\square$

Thm (Abyah)

Any object X satisfying bichain condition
 can be decomposed into a finite direct sum
 of indecomposables w/ local endo rings
 (i.e. $X = \bigoplus_{i=1}^r X_i$ w/ X_i - indec $\Rightarrow \text{End}_A(X_i)$ - local)

Proof: Suppose X doesn't decompose like that.

But $X = X_1 \oplus Y_1 \Rightarrow X_{n+1} = X_n \oplus Y_n$ then

$X_n \twoheadrightarrow X_{n+1} \hookrightarrow X_n$ $\alpha_n \beta_n = \text{id}_{X_n}$. Since X

satisfies bichain condition, X_n must terminate

$\Rightarrow$ decomposition

□

Prop 1.13 A ab fl ep then proj covers of
 simple objects exist.

Proof: $L \in A$ simple $\exists$ proj $P \twoheadrightarrow L$

WLOG P indec $\Rightarrow$ by lifting lemma $P \rightarrow P$ nilp/iso

$$\begin{array}{ccc} P & \xrightarrow{\varphi} & L \\ \alpha \uparrow & \nearrow & \\ Q & \neq 0 & \end{array}$$

epi since L -simple $\Rightarrow$

$$\begin{array}{ccc} & P & \\ \swarrow & \downarrow & \\ Q & \xrightarrow{\varphi \circ \alpha} & L \end{array}$$

$\varphi \circ \alpha$ not nilp $\Rightarrow$ iso

$\Rightarrow \alpha$ epi

$\Rightarrow P \twoheadrightarrow L$ is a proj cover of L

□

Ex (careful! some inaccuracies in copying)

Consider the quiver Γ (just imagine an oriented graph)

$$\begin{matrix} & 1 & & 2 & & 3 \\ & \bullet & \xrightarrow{\quad} & \bullet & \xrightarrow{\quad} & \bullet \end{matrix}$$

Let $\mathcal{A}$ - cat of fd repr of Γ

= cat of functors $\Gamma \rightarrow \text{Vec}_k = k\Gamma^{\text{op}}\text{-mod}$

$\downarrow$
"path algebra of Γ "

Simple objects

Proj covers

$$L_1 \quad \mathbb{C} \rightarrow 0 \rightarrow 0$$

$$P_1 \quad \mathbb{C} \rightarrow \mathbb{C} \rightarrow \mathbb{C} \quad k\Gamma e_1$$

$$L_2 \quad 0 \rightarrow \mathbb{C} \rightarrow 0$$

$$P_2 \quad 0 \rightarrow \mathbb{C} \rightarrow \mathbb{C} \quad k\Gamma e_2$$

$$L_3 \quad 0 \rightarrow 0 \rightarrow \mathbb{C}$$

$$P_3 \quad 0 \rightarrow 0 \rightarrow \mathbb{C} \quad k\Gamma e_3$$

Note: $\dim \text{Hom}_{k\Gamma}(P_i, P_j)$

$$= \dim \text{Hom}_{k\Gamma}(k\Gamma e_i, k\Gamma e_j)$$

$$= \dim e_i k\Gamma e_j$$

$$\begin{array}{ccc} f & & \bullet \times \\ \downarrow & & \uparrow \\ f(e_i) & & \times \end{array}$$

$$= \# \text{ of paths } i \rightarrow j$$

$$= [P_j : L_i] \iff \text{multiplicity of } L_i$$

$\uparrow$
 $\{$

we will see this later